

Exam

Numerical Programming I

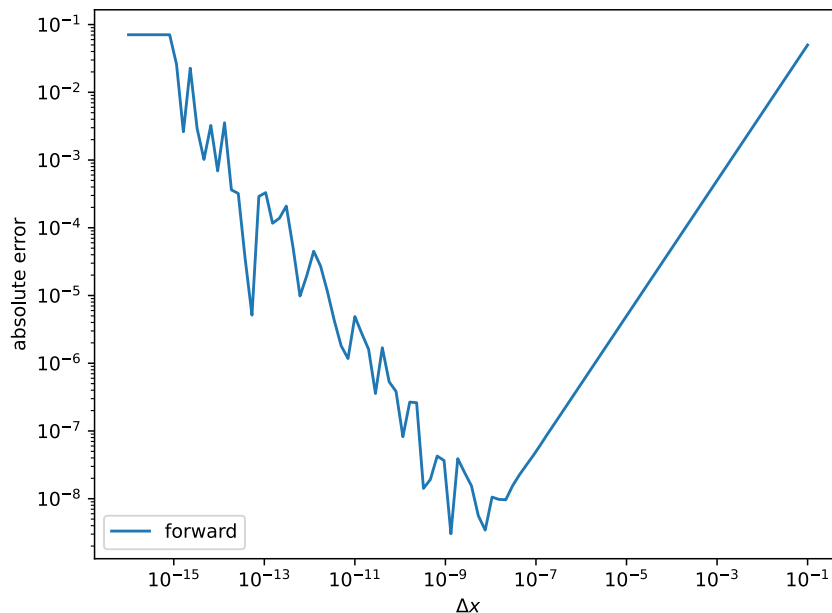
February 6, 2026

Name: _____

Question:	1	2	3	4	Total
Points:	5	5	9	0	19
Score:					

Mixed Short Questions

1. (a) What does $(1.0+10^{-17})-1.0$ evaluate to in double-precision floating point arithmetic (with machine precision $\epsilon_{\text{mach}} \sim 10^{-16}$)?
- ☐ 10^{-17} is correctly calculated.
 - ☐ 10^{-17} cannot be represented in double-precision, the expression results in undefined behavior.
 - ☐ The expression evaluates to zero, the mantissa of 1.0 does not have enough digits to capture 10^{-17} .
- 2 (b) Consider a differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$. We have written a program which approximates $f'(x)$ using the forward difference $\frac{f(x+\Delta x)-f(x)}{\Delta x}$. Below, the absolute error $\left| f'(x) - \frac{f(x+\Delta x)-f(x)}{\Delta x} \right|$ over Δx is plotted. Mark the region where the floating point error is dominant and the region where the truncation error is dominant.



- 2 (c) Does gradient descent always converge to the **global** minimum? Answer with yes or no and give a short justification.

Second order derivative approximation

2. For a twice differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ we would like to find a finite difference approximation for the second derivative $f''(x)$.

2

- (a) Approximate $f(x + \Delta x)$ and $f(x - \Delta x)$ by Taylor expansion up to (including) third order around x .

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- (b) Using these approximations find an approximation for $f''(x)$ based on $f(x)$, $f(x + \Delta x)$ and $f(x - \Delta x)$. What is the order of the error term? *Hint:* You need to combine both Taylor expansions such that the first and third order derivatives cancel.

1

- (c) For $f(x) = x^2$ and $\Delta x = 0.1$ approximate $f''(0)$ and compare your approximation to the analytical value.



Convergence of Newton's root finding method

3. In the lecture we have stated:

If g is continuously differentiable in a neighborhood around a root x^* with nonzero derivative at x^* , then there exists a neighborhood around x^* where Newton's iteration converges to x^* for all starting values.

Now consider the function

$$g(x) = x(1 + x^2)^{-\frac{1}{2}} \quad (1)$$

- 1 (a) What is the root x^* of g where $g(x^*) = 0$?

- 2 (b) Show that the derivative of g is $g'(x) = (1 + x^2)^{-\frac{3}{2}}$. *Hint:* Calculate the derivative by product rule and factor out $(1 + x^2)^{-\frac{3}{2}}$.

- 1 (c) Calculate $g'(x^*)$. Is there a neighborhood around x^* where Newton's iteration converges to x^* for all starting values inside that neighborhood?

- 2 (d) Show that the Newton step

$$x_{k+1} = x_k - \frac{g(x_k)}{g'(x_k)} \quad (2)$$

is given by $x_{k+1} = -x_k^3$.

- 3 (e) Consider the following cases for the initialization of the Newton iteration

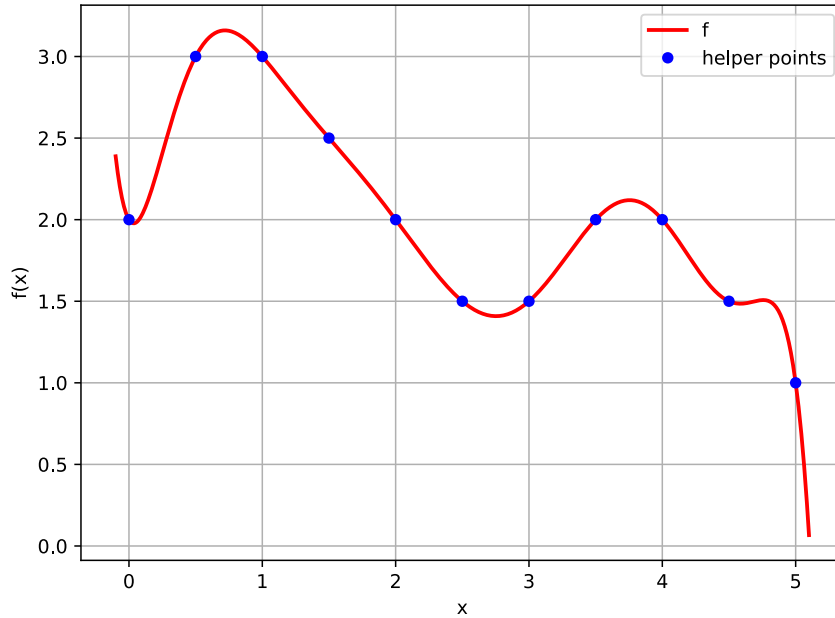
- $x_0 = 0.5$
- $x_0 = -1.0$
- $x_0 = 2.0$

and analyze if the iteration converges, diverges or oscillates.



(Bonus) Numerical integration by hand

4. Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ plotted below.



We want to approximate the integral

$$\int_0^5 f(x) dx \quad (3)$$

Here, assume the subdivision

$$\int_0^5 f(x) dx = \sum_{i=0}^{N-1} \int_{x_i}^{x_{i+1}} f(x) dx \quad (4)$$

with $N = 5$ and $x_0 = 0.0, x_1 = 1.0, x_2 = 2.0, x_3 = 3.0, x_4 = 4.0, x_5 = 5.0$.

And consider the following *quadrature rules*:

- midpoint rule:

$$\int_{x_i}^{x_{i+1}} f(x) dx = \underbrace{(x_{i+1} - x_i) f\left(\frac{x_i + x_{i+1}}{2}\right)}_{=: M_i} + \mathcal{O}((x_{i+1} - x_i)^3) \quad (5)$$

- trapezoidal rule:

$$\int_{x_i}^{x_{i+1}} f(x) dx = \underbrace{(x_{i+1} - x_i) \left(\frac{f(x_i) + f(x_{i+1})}{2} \right)}_{=: T_i} + \mathcal{O}((x_{i+1} - x_i)^3) \quad (6)$$

2 (bonus)

(a) Approximate $\int_0^5 f(x) dx$ using the midpoint rule as given above.



2 (bonus)

(b) Approximate $\int_0^5 f(x) dx$ using the trapezoidal rule as given above.